Observer-Patch Holography Cosmological Vacuum and Structure Formation: Finite Transfer Conditions, Fluctuation Ensembles, and Proto-Objects

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Abstract

We specify finite quotient ensembles and transfer conditions for Observer-Patch Holography (OPH) cosmology. A declared quotient-intrinsic action and base measure determine a finite state, while reversible Euclidean data, reflection positivity and compatible refinement support its vacuum interpretation. On a supplied Whitney action, controlled real-field trajectories and a neutral interacting Hilbert space provide explicit matter and state constructions, and we state the further source and transfer conditions that connect such objects to primordial seeds, observer-facing worldlines and large-scale structure observables. The source screens of the executable observer construction test these conditions: their native density law has a reflection-positive reconstruction whose generator is the native continuum operator, and the full return reference approaches the normalized trace, so no quantum vacuum is identified. On the sourced causal order the expansion history is a record density. The capacity dictionary $\Lambda\ell_P^2 = 3\pi/N$, the de Sitter horizon entropy read backwards, returns, as diagnostics of two target-informed closure candidates, the cosmological constant $+0.62$ and $+0.38$ percent above the Planck central value, while the candidates sit -2.64 and -2.81 standard deviations below the DESI DR2 display of the same coordinate; under a Gaussian moment diagnostic the fixed-capacity equation of state $w = -1$ sits 2.6 to 3.9 standard deviations from the DESI DR2 posterior means.

1 Scope

This technical companion states the vacuum and fluctuation boundary for finite-source cosmology, screen microphysics, the compact results, and finite-patch simulations. The executable observer construction runs native repair, record-controlled feedback and local event clocks on one twelve-port carrier [5, 3]. Its refining source screens give a positive continuum operator of spectral dimension two [6]; Section 4 states why its native reference identifies no quantum vacuum. That construction does not supply the source Euclidean slab, expanding background or transfer conditions stated below. The simulator source and its canonical public data archives are cited in [4]. The large finite repair archive is a diagnostic that passes neither the physical-CMB nor the production-gravity criterion; it does not supply the source and transfer receipts required below.

The spacetime construction [1] gives a controlled flat $1+3$ causal-order and count-volume limit for conservative Fibonacci records with a specified population, complete local neighbour reads and

layer duration. An observer with complete authenticated ancestry and interval anchors can read $(|I|/|J|)^{1/4}$ relative to a nondegenerate interval J in the same history. For interior timelike diamonds this tends to the proper-time ratio; the readout uses no coordinates, timestamps or density. Its reference fixes relative units, and the law underlying the limit is supplied.

The vacuum term of the background has one dictionary on the fixed-capacity branch, $\Lambda\ell_P^2 = 3\pi/N$. Reading N as de Sitter horizon entropy is a premise, and with it the dictionary is that entropy, $N = 3\pi/(\Lambda\ell_P^2)$ in nats, inverted: it holds by definition and fixes no independent value of N. The comparison below therefore tests candidate formulas for N, not the dictionary. On the native source, regional reduced entropy grows with port count and mutual information per crossing seam tends to zero, so a whole-screen area reading requires a region map that the executable construction does not supply [10]. The two closure candidates for the capacity, 3.2921×10^{122} and 3.3001×10^{122}, return $\Lambda = 1.0959 \times 10^{-52}$ and $1.0933 \times 10^{-52}\,\mathrm{m}^{-2}$ against the Planck-central $1.0891 \times 10^{-52}\,\mathrm{m}^{-2}$, residuals of $+0.62$ and $+0.38$ percent, and sit 2.64 and 2.81 posterior standard deviations below the DESI DR2 with CMB display $(2.968 \pm 0.040) \times 10^{-122}$; both brackets are exact rational inequalities machine-checked in the Lean module `CosmologyLedgerBrackets`. Both candidates were exposed after the data and are target-informed declared maps, so the residual is a diagnostic of a frozen formula and carries no predictive weight; a relative capacity residual δ gives a relative constant residual $-\delta/(1+\delta)$, so the large capacity explains the small scale and not the closeness. On the same branch a constant capacity gives $w = -1$ exactly and a monotone capacity gives $w \geq -1$; the Gaussian moment diagnostic of the official DESI DR2 chains places $(-1, 0)$ between 2.6σ and 3.9σ from the four default posterior means, with monotone-branch posterior fractions between 0.007 and 0.11 percent, so the fixed branch is under tension on seen data. The conditional FZ-13 protocol was frozen on 2026-10-04 with register status `frozen_stamped_upgrade_pending`; only its complete protocol and anchoring requirements define the eligible comparison. The fixed-capacity point coincides with ΛCDM, so its survival earns no distinctive OPH support. FZ-01, frozen on 2026-07-17, registers the capacity band $N \in [3.30836, 3.31137] \times 10^{122}$ and the point $(w_0, w_a) = (-1, 0)$ without a registered verdict; both closure candidates lie below that band [12]. The cosmology postdiction ledger released with these papers records these rows with their classes.

On the rank-three source Gram space with a real axis adjoined, a nonzero closed convex pointed cone is future or past Lorentzian if the proper icosahedral rotations and all rapidities of one boost axis preserve it. Time orientation selects the future cone. This premise must concern possible influence on histories, interventions and clock readouts, beyond an algebraic frame action. The executable carrier supplies one such influence bound: for a fixed seam schedule and shared local coins, dependence crosses at most one seam per ordered repair event, which gives no laboratory speed or continuum light cone [11]. A cosmological vacuum requires the additional state, expanding geometry, physical clock calibration and screen-to-volume transfer conditions below. Smooth Einstein promotion requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic.

2 Matter and States on a Supplied Spatial Domain

The Whitney construction on the fixed tetrahedral cone gives a controlled matter continuum in a specified sector of one charged scalar action. For $m^2 > 0$, $g \geq 0$, a supplied $u \in C^2([0,T]; H^2(\Omega))$ solving the real cubic Klein–Gordon equation with natural Neumann boundary, and Ritz-projected initial data, uniform conforming refinements give

$$\sup_{0 \leq t \leq T} \left(\|u_n - u\|_{H^1} + \|\dot{u}_n - u_t\|_{L^2} \right) \leq C_T n^{-1}.$$

The real, zero-current solutions are invariant solutions of the full charged action. The proof uses a fixed shape-regular refinement family, the conserved nonlinear energy and a Ritz error estimate, as proved in *Observers Are All You Need* [2].

On the original finite mesh with nonzero coupling e, eliminating mean-zero gauge directions gives a complete kinetic metric γ on $\mathcal{Q} \cong \mathbb{R}^{30} \times \mathbb{C}^{13}$. With supplied $\hbar > 0$ and Laplace–Beltrami ordering, the neutral subspace of $L^2(\mathcal{Q}, d\mathrm{vol}_\gamma)$ has a unique nonnegative self-adjoint Hamiltonian and unitary dynamics. Normalized Gaussian half-density states lie in its operator domain and have explicit initial matter moments [2]. These are states of a finite spatial discretization, not a derived cosmological ground state or a continuum quantum field limit.

For the same autonomous action, the Jacobi duration

$$d\tau = \sqrt{\frac{G_q(dq, dq)}{2(E - V(q))}}$$

recovers action time on regular nonturning solutions at supplied energy E, using the full coupled kinetic metric or its gauge reduction [2]. Its configuration-based reconstruction supplies an internal model clock. A cosmological interpretation additionally requires observer-selected geometry, physical preparation and units, and a compatible expanding background. Those attachments are separate from these matter and state theorems.

The golden source population of the spacetime construction carries a second matter and state construction on the same population that the causal order uses. Its positive source-metric scalar action has a finite-time continuum limit with an explicit detector error bound, a canonical quantization whose quadratic Hamiltonian has a unique normalized ground state, the original action vacuum, and a bounded probability comparison between the split evolution of that vacuum with a coherent displacement and the continuous evolution of the same finite action, with all sixty-four modes retained at $q = 5$. The joint spatial and temporal refinement of one compact detector is certified at $q = 233$, with total error below 0.0501 against a resolved response above 0.1377. This is a vacuum state of a finite scalar action on the source population, with the population, action and tick supplied. It is neither a reflection-positive Euclidean vacuum of a cosmological release law nor a source-instantiated triple $(J_r, V_r, a_{t,r})$ in the sense of the next section, and it carries no expanding background.

On the same family the expansion history has an exact finite form. The layered read law in comoving coordinates is conformally invariant, so the flat causal order is the causal order of every spatially flat expanding history; the scale factor enters only as the record-density weight a^4 per comoving cell and conformal tick. The count clock approaches the proper-time ratio for shrinking diamonds; a fixed finite window can retain a continuum bias. The redshift between two epochs is the fourth root of a record-count ratio for congruent comoving windows whose scale factor is constant within each, or in the shrinking-window limit of a continuous profile. The finite-source cosmology paper states and proves these statements, with their Lean module. The record-production profile is a supplied datum; the vacuum and fluctuation conditions below concern the field on that background and do not select the profile.

3 Finite Quotient Ensemble Theorem

Finite quotient ensemble theorem. Fix a finite regulator r. The physical presentation space is Σ_r, the presentation redundancy groupoid is Γ_r, and the finite physical quotient is

$$Q_r = \Sigma_r / \Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

At least one action value must be finite, so that $Z_r > 0$. For a law on settled configurations only, restrict this construction to N_r, or set $S_r = +\infty$ outside N_r. A separately specified prior ν_r on Q_r instead induces the settled law $\mu_r^{\mathrm{nf}} = (n_r)_{\#}\nu_r$. These constructions agree only if

$$\sum_{x:\, n_r(x)=q} \nu_r(x) = \mu_r(q) \qquad (q \in N_r)$$

and μ_r has no mass outside N_r. For example, a retraction $n(0) = 0, n(1) = n(2) = 1$ sends the uniform law on three states to weights $(1/3, 2/3)$, whereas uniform counting on its two normal forms gives $(1/2, 1/2)$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to Q_{nf} is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#}\mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X. If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q, positive m, and quotient observables $F_1, \ldots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r, self-adjoint constraints and a feasible set containing a faithful state,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

If the feasible states lie in a proper support projection P, first restrict to their joint support. The same multiplier argument uses $P(\log \sigma_r)P$ and $PF_{r,a}P$, exponentiated and normalized on $P\mathcal{H}$; replacing the reference by $P\sigma_r P$ before taking its logarithm is not valid without a commutation condition. An empty feasible set has no selected state. The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse maps, with $c_{tr} = c_{sr} \circ c_{ts}$ for $t \succeq s \succeq r$. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' :\, c_{sr}(q')=q} m_s(q')e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q. Then

$$(c_{sr})_{\#}\mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \left\| (c_{k+1,k})_{\#}\mu_{k+1} - \mu_k \right\|_{\mathrm{TV}},$$

then

$$\left\| (c_{nr})_{\#}\mu_n - \mu_r \right\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} \left[\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r \right]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε, the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#}\mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x :\, \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\widetilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\text{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant on the orthogonal complement of the common fixed space $\mathcal{F}_r = \bigcap_C \operatorname{Ran} E_C$:

$$\kappa_r = \inf_{0 \neq f \perp \mathcal{F}_r} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If every $c_C \geq \gamma_* > 0$, then

$$L_{\text{rep}} \geq \gamma_* \kappa_r (I - P_0),$$

where P_0 projects onto $\mathcal{F}_r$. In finite dimension, $\kappa_r > 0$ whenever $\mathcal{F}_r^{\perp} \neq \{0\}$, since zero Dirichlet energy would put f in every fixed range. If all states are fixed, the displayed operator inequality is vacuous. Replacing $f \perp \mathcal{F}_r$ by $f \perp 1$ requires the additional ergodicity condition $\mathcal{F}_r = \operatorname{span}\{1\}$; a conserved nonconstant record otherwise makes that infimum zero. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt. Multiplying all rates by $s > 0$ rescales the generator and gap by s, while leaving the fixed space, stationarity and detailed balance unchanged. Numerical recognition therefore tests identities after removing a common rate scale and restores that scale to reported gaps. An unresolved positive gap is a precision limitation, not evidence that changing clock units changed the equilibrium.

Finite evidence accuracy. For bounded coarse observables O, if

$$\|\widehat{\mu}_s - \mu_s\|_{\text{TV}} \leq \epsilon_{\text{samp}}$$

and the refinement defects sum to ϵ_{ref}, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \leq 2\|O\|_\infty(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Physical-vacuum requirement. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q,q') = J_r(q',q) \geq 0$, local potential V_r, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q,q')(f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac}. A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family. On the native source screens the actual density law has such a reflection-positive reconstruction along one common refining subsequence; its one-particle generator is the native continuum operator A rather than $\sqrt{A}$, and it supplies no coherent preparation, relativistic vacuum or Lorentz symmetry.[1]

Primordial and cosmological prediction firewall. A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance. On the native source screens the continuum energy is not boundedly congruent to the static boundary action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ while the native eigenvalues grow like n, and internal rank-three carrier modes vanish from bounded-energy families; a radial lift needs a radial observation algebra that the native energy does not supply.[2]

Interpretation classes and required receipts. Every ensemble-facing run records its ensemble id, interpretation class, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2053/model/observer_time_fields/REFINEMENT.md

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2053/model/physical_space_time/OCCUPATION_FIELD.md

policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are

seed, proposal, or repair noise,
conventional reference ensemble,
OPH-native quotient ensemble,
OPH physical vacuum,
OPH primordial field,
observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

Native source screens. On the native source screens the actual density law has the reflection-positive reconstruction that a continuum interpretation requires, along one common refining subsequence; it supplies no coherent preparation, relativistic vacuum or Lorentz symmetry [7] (Section 4). The same continuum energy is not boundedly congruent to the static boundary action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ while the native eigenvalues grow like n, and internal rank-three carrier modes vanish from bounded-energy families; a radial lift needs a radial observation algebra that the native energy does not supply [6].

4 Vacuum Claim Boundary

A simulated field is classified as seed noise, repair jitter, or a conventional reference ensemble unless it satisfies the finite quotient ensemble conditions. An OPH-native vacuum requires source data of the form

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

with J_r, V_r, and $a_{t,r}$ derived from OPH source structure rather than from a target law or a desired visual spectrum. Directed accepted repair moves cannot be reused as reversible Euclidean conductances. The native seam rule differs: each unread repair has the law of a fair identity-or-endpoint swap, so its configuration transfer K is symmetric and positive, and the actual native density law has a reflection-positive Gaussian reconstruction. The obstruction on the native source lies elsewhere: the reconstructed one-particle generator is the continuum operator A, not $\sqrt{A}$, and the full configuration return reference built from K approaches the normalized trace at fixed modular parameter and gives the density-wave sector vanishing weight, so it identifies no quantum vacuum [7].

4.1 Vacuum and initial-state readouts on the effective product

The active product construction has a different, conditional state contract. With its positive spatial operator B, canonical fields and supplied inverse temperature b, its ordered free-field two-point function is

$$C_b(t) = \frac{\coth(b\sqrt{B}/2)\cos(t\sqrt{B}) - i\sin(t\sqrt{B})}{2\sqrt{B}}.$$

The limit $b \to \infty$ gives the static ground-state covariance $(2\sqrt{B})^{-1}$; finite b changes fluctuations while preserving the same commutator and response. These are joint observer-readout predictions of the declared free-field family, not the passive native density law or the interacting field's covariance [7]. The radial apparatus, state preparation and physical clock map remain inputs.

On the finite interacting work algebra $\mathcal{A} = M_8(\mathbb{C}) \oplus M_8(\mathbb{C})$, restricted thermal equilibrium leaves a central sector weight:

$$\rho_w = w\frac{e^{-bH_+}}{Z_+} \oplus (1-w)\frac{e^{-bH_-}}{Z_-}, \qquad 0 < w < 1.$$

Every such state has the same restricted modular flow. Cooling each block does not select w. The full Gibbs preparation additionally chooses $w = Z_+/(Z_+ + Z_-)$; its global ground-state limit is therefore a stronger state choice. Admitted record statistics can test sector populations, but do not identify an unobserved purification [8].

A direct cosmological identification has a discriminating control. In the flat massless limit, with $\hbar = c = 1$ and $\langle \phi_{\mathbf{k}} \phi_{\mathbf{k}'} \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') P_\phi(k)$, the static vacuum gives

$$P_\phi(k) = \frac{1}{2k}, \qquad \Delta^2_\phi(k) = \frac{k^3 P_\phi(k)}{2\pi^2} = \frac{k^2}{4\pi^2}.$$

Thus a scale-independent identification of this field with primordial curvature does not reproduce the nearly scale-invariant curvature spectrum inferred from CMB data [9]. This flat-limit control is not a Fourier-spectrum claim for the bounded native radial product. An expanding evolution and an operational curvature map must supply the transfer before the state can predict cosmological observations. The initial-state target is consequently a law for accessible earlier-state correlations and their later records, not an assumed observable absolute beginning of the Universe.

5 Structure-Formation Requirements

Object	Required source-side label
Fluctuating screen field	Seed noise or a conventional reference ensemble unless the quotient ensemble conditions hold.
OPH vacuum field	Source Euclidean slab, ground-state, Doob, reflection-positivity, and transfer-refinement conditions.
Proto-object worldlines	Diagnostic until production matter/particle and observer-facing object population conditions hold.
Growth seeds	Primordial bridge, physical mode basis, radial-null, positivity, forward-projection, and source-dilation or tomography conditions.
Large-scale structure comparison	Diagnostic; source, transfer, and official likelihood/data contracts are not supplied.

6 Numerical Acceptance Conditions

The finite-patch calculation reports the following five conditions separately:

- a tracially pointed quotient;
- a source-derived Euclidean slab;
- transfer stability under refinement;
- schedule invariance of the stationary law;
- pathwise invariance under partition changes.

A failed condition is part of the scientific result and excludes the physical vacuum classification for a reference baseline or rendering field.

7 Unsupplied Premises

The cosmological source construction does not supply the reversible conductances J_r, potential V_r, or slab thickness $a_{t,r}$ required above. A source-instantiated reflection-positive vacuum, or an equivalent reconstruction on this cosmological branch, is not supplied. The bridge from a source-derived vacuum to primordial growth seeds without cosmic microwave background or large-scale structure data is not supplied. A physical proto-object or worldline requires conditions distinct from those used for production particle claims.

References

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